Peng 2019 PDAC — recapitulation summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → PASS

16/16 samples (≥100 cells) intermix on one continuous manifold; no per-sample islands (0 detached cells removed). Median iLISI 4.33 = 57% of the proportional-mixing maximum (7.53); neighbour graph is a single connected component.

Gate 2 — module geography → NO-GO

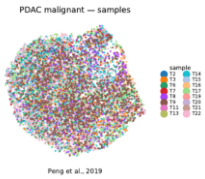
The three module-high territories sit at three separate UMAP corners (triangle, angles 44/95/42° — not a Gast/pSom body flowing to a Crest terminus). Gast & pSom are spatially mutually exclusive: joint-high 0.8% vs 6.2% expected if independent (enrichment 0.13, <1), so there is no co-expression hub and no co-varying body. Along the rooted axis Gast (ρ=−0.56) and pSom (ρ=+0.11) move in opposite directions and Crest is flat (ρ=+0.10; high-decile pseudotime 0.70, mid-axis, not terminal). The anti-correlation holds at every clustering scale (sub-region sweep). Neither GO pattern — branching hub nor non-branching Gast/pSom overlap — is present.

Gate 3 — hub cross-repression → NA

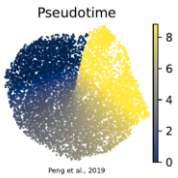
No bifurcation (single non-branching axis); Gate 3 is not evaluated.

STEP FIGURES

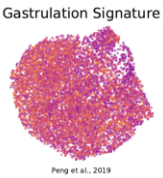
Gate 1: integration



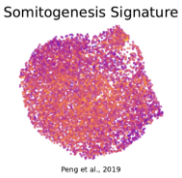
Trajectory / root



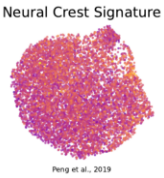
Gast



pSom



Crest



EMT-TFs



Dataset. Pancreatic ductal adenocarcinoma (Peng et al., 2019), 10x — 16 patients / 27,953 cells (master scope); malignant only: 7,945 load, all 7,945 enter the trajectory (one Leiden lobe, no subtypes).

Verdict. ✗ Pattern does not reproduce — Gate 2 NO-GO. The embryonic modules are present (pSom mean +0.10, strongest in the atlas) but resolved into three mutually-exclusive territories, not a Gast/pSom hub or co-varying body; no bifurcation, no terminal Crest. The smoothed Gast~pSom = −0.41 reflects two separate corners, not cross-repression within a shared hub.

Parameter choices

Min cells/sample	≥ 100 (keeps 16/16)	**Integration**	Harmony on sample, θ=10, 30 it
Island removal	UMAP radius-graph (−0 cells)	**Trajectory**	ElPiGraph curve N=20 μ=50 λ=0.3
Root	tip max(Gast+pSom−Crest)	**Geometry**	non-branching blob; 1 lobe, no subtypes